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Studierichting: Informatica

Oefeningenreeks 1C nr. 31.14

$$\begin{array}{c|cc}
 \left(\frac{1-3i}{2-i}\right)^8 & & \\
 \hline
 r & \frac{1-3i}{\sqrt{10}} & \frac{2-i}{\sqrt{5}} \\
 \tan \theta & -3 & -\frac{1}{2} \\
 \arg. z & \text{Bgtan}(-3) & \text{Bgtan}\left(-\frac{1}{2}\right)
 \end{array}$$

$$\Rightarrow \left(\frac{1-3i}{2-i}\right)^8 = \left(\frac{\sqrt{10} \text{cis Bgtan}(-3)}{\sqrt{5} \text{cis Bgtan}\left(-\frac{1}{2}\right)}\right)^8 = \left(\sqrt{2} \text{cis Bgtan}\left(\frac{-3 + \frac{1}{2}}{1 + (-3)\left(-\frac{1}{2}\right)}\right)\right)^8$$

$$= (\sqrt{2} \text{cis}(\text{Bgtan}(-1)))^8 = \left(\sqrt{2} \text{cis}\left(\frac{3\pi}{4}\right)\right)^8 = 16 \text{cis}\left(\frac{24\pi}{4}\right) = 16 \text{cis}(6\pi) = 16$$

References